Design of Sustainable Blood Bank Management System for Biomedical Applications

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Abstract- Modern healthcare systems are built on the compassionate act of blood donation, which entails voluntarily giving a part of oneself to save lives. An overview of the significance of blood donation, its role in medical emergencies, and its long-lasting effects on people and communities. Blood donation has effects that extend beyond the immediate medical setting. Communities are strengthened when members band together to support one another, frequently in difficult circumstances. Donating encourages a feeling of fulfillment and empowerment, strengthening the conviction that people can genuinely improve the lives of others. Providing a steady supply of blood for different medical procedures, surgeries, and emergencies, blood donation acts as a lifeline. The selfless and charitable nature of blood donation reflects the social values of empathy and cooperation. For patients in need, such as accident victims, people undergoing complicated surgeries, or people with chronic illnesses, donated blood that has been carefully screened and processed is a vital resource. In conclusion, giving blood is a symbol of humanity's compassion and empathetic nature. It cuts across borders, cultures, and backgrounds, capturing the spirit of unity and support. By giving blood, people support not only the health and happiness of the recipients but also the overall resilience of society at a time of need.

I. Introduction

A bank for blood is a convenience that guarantees the appropriate blood storage after it has been securely collected at blood donation concentration camps. A blood center is a facility where donated blood is gathered and stored in preparation for future blood transfusions.The word "blood bank" also describes a facility for the safe storage and upkeep of blood through appropriate testing to ensure that complications from transfusions can be avoided. Hospital blood banks are responsible for the safe storage and retrieval of blood, but the procedure is not as straightforward as it might seem. Before administering blood, the medical facility must first inspect both the recipient and the blood themselves to make sure a smoother blood transfusion. The management of essential resources like blood has entered a new phase of efficiency and effectiveness in an age marked by digital connectivity and technological advancement.

The creation of an online donor blood bank administration system has greatly improved the complicated processes of blood donation, storage, and distribution. This ground-breaking system addresses historical issues and transforms the blood management landscape by utilizing the power of the internet to build a centralized platform that seamlessly connects donors, recipients, and healthcare organizations.

Logistical challenges, poor communication, and delays that might affect patient care have historically characterized blood donation and distribution. By providing a thorough digital interface, the Online Blood Bank Management System, which recently emerged, directly addresses these concerns. This interface makes it simple to register donors, monitor inventory in real-time, and receive alerts on time, creating a cohesive ecosystem that puts a priority on quick access to compatible and safe blood.

It is an important component of technology for this system's streamlined blood collection, storage, and distribution processes. The blood supply chain as a whole is more efficient, accurate, and secure thanks to the blood facility managing system by incorporating technology into blood banking operations. This results in better patient care, quicker emergency response times, and an overall improvement in public.

II Literature Survey

A Study on Blood Bank Management [1]

This application's main objective is to outline a bank for blood's which covers entire operation in this paper. The application must maintain several thousand of records. Additionally, search should be quick so that users can find the data they require right away. It is excellent for getting information. In order to increase the accuracy and effectiveness of donated blood and administration systems. They assess the condition of blood bank facilities currently in use, analyse their advantages and disadvantages, and suggest a new system that incorporates cutting-edge technologies.

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Blood Bank Management System.[2]
This paper discusses the development of an emergency-useable cloud-based blood center organization system. It provides an online database of medical facilities, blood donation centres and contributors in addition to a sophisticated blood handling system.

Their principal objective is to create a network that connects all blood bank medical centres contributors, and blood the financial institution clinics in order to gather, store, and analyse various data about bloodstream and the well-being of individuals. Users can only access this database's information if they are logged in, and the system's user-friendly mobile application makes registering for an account simple.

Smart Blood Query[3]
The mobile system Smart Blood Query (SBQ) chooses the top five donors from the database and contacts them for assistance. If volunteers don't respond, the requester is given information about a nearby blood bank. It would benefit the person making the request. This paper presents a location-aware mobile device-based blood supply recruitment, data retrieval, and control system to ensure blood quality and improve operation management efficiency.

Smart Phones to the Rescue[4]
The article discusses the Project for an Online BloodCenter, in which the technique also identifies a blood donor who lives nearby in the database and sends the requester their contact information. The VBB project uses mobile devices to build a widespread system that gives individuals access to current data about those who donate blood in the area in Delhi. With no specialized hardware required, subscribers can access the service whenever they want thanks to the Virtual Blood Bank project.

The Optimization of Blood Donor Information and Management [5]
Priya et al. [5] introduced a GIS-equipped Android mobile application that creates a secure, efficient, and effective information management system. The suggested remedy finds phony donors and app users who abuse user information. Additionally, various levels of blood quality checking are now available and it is necessary for the patient's safety.The managers are able to view all the information regarding the system used by the blood bank and quickly modify the details about funding sources, receivers, and clients using the enhanced online tool that they demonstrate. The proposed work also incorporates the push technique with security to protect donors' names and addresses in online applications where it might be misused by third parties.

A Study on Blood Bank Management System[6]
For this website, they developed a functional data management structure that manages the records of contributors and clients. Only authorized Blood Bank employees can view and edit the records by logging in with a secret login password thanks to a security layer.They have created an app that encourages a healthy community by reminding donors when they are once again eligible, providing the places of blood drive centres in the area, making it simpler for patients to inquire about blood, etc. IOT is utilized to link the server and application. and to communicate between applications. This data exchange and collection process is made more effective with the aid of IOT.

Development of a Blood Bank Management System [7]
To meet the needs of the Sultanah Nur Zahirah Hospital (HSNZ), a web-based managing system was developed. Various hospitals might handle them in a various manner. Even though the staff members are not present at the HSNZ, the data entered will be showed to the different blood donation staffs, such as nurses, so that everyone can be up-to-date of the timetable for the blood contribution event. The Rational Unified Process (RUP) was employed in the construction of this system.

Blood-bank-donor-management-system[8]
This study uses a online webpage application called the bank for blood and Donor Managing System. People who enjoy giving blood register in the app.

People who want blood look for people in the city who belong to the same blood group. If he locates a donor in his city, he receives all of the donor's information; if he is unable to locate a donor, he is provided with the contact information and address of the Life-Saving Contact Persons in his city.

Blood-bank-management-system-in-php[9]
There are two different types of logins for this project that is, Admin and User. A user can view the dashboard, request blood, and donate blood after logging in. The administrator has complete control over the system and can add and manage states, cities, and members. Additionally, he or she can maintain Active and Nonactive donors from the admin panel. In a nutshell, the system facilitates online blood donations and requests, donor management, and blood request details. This project's design is straightforward, making it easy for the user to comprehend, use, and navigate.

III PROPOSED WORK

An system for managing blood banks using online has been developed over a number of stages using front-end technologies like HTML, CSS, and React.js for interaction between users and back-end tools like Node.js and MongoDB over data processing and storage [9 -13].

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The following detailed process will guide about the development process:

Front-End Development (HTML, CSS and React.js):

Step 1: Requirements Analysis

By consulting with stakeholders, such as managers of blood banks and health care providers, to comprehend the prerequisites for the administration system of the donated blood bank and to ascertain the crucial traits and qualities.

Step 2: Design the User Interface (UI)

To visualize the layout and structure of the user interface, create wireframes and design mockups. Use responsive design to make sure the system is compatible with desktop, tablet, and mobile devices

Step 3: Creating the Front-End

Using HTML to build the foundation for web pages and style the pages using CSS to give them a pleasing appearance and using JavaScript to make the user interface more interactive. This includes client-side functionality, real-time updates, and form validation. Instead, to create a dynamic and interactive user interface, we use React.js, which comes with built-in HTML and CSS. Reusable UI elements can be created using React components.

Back-End Development (Node.js, MongoDB, SQL):

Step 4: Database Design and Setup

For designing the database schema to store data such as user information, donor details, recipient records, blood inventory, donation history, and more, select the MongoDB database system, which enables a powerful document-based querying system which allows dynamic document queries. MongoDB is a document database where various types of documents are stored in a single set of documents. One document could vary from another in terms of the number of fields, content, and document size.

Step 5: Server-Side Development

Using Node.js as a server-side runtime environment, we apply authentication and authorization to users using Passport.js and other libraries to manage HTTP requests and responses. Create protected login and registration procedures and define user roles (administrator, donor, recipient, etc.).

Step 6: Implement Business Logic

Construct the core elements of a web-based blood banking system using Node.js and MongoDB, includes donor and recipient management (registration, data storage), and accessibility of blood tracking (recording blood types, quantities, date of expiration, and low stock alerts). Managing donation camps (plan and organize camps, keep track of locations).Notifications (notify recipients, remind donors).Security (use mechanisms for data validation, encryption, and protection).

Step 7: API Development

Create, Read, Update, Delete operations should be implemented for data management, and RESTful APIs should be developed using Node.js to improve front-toback communication.

Step 8: Examining and Assurance of Quality Execute

acceptance tests for users , tests for integration, and testing for functionality on the system's front end and back end in order to find and fix bugs and other problems.

Step 9: Documentation

Provide comprehensive documentation for users, administrators, and developers. Include user manuals, API documentation, system architecture documentation, and code documentation.

Step 10: Deployment and Hosting

For deployment, choose a reputable hosting environment set up your Node.js application for maximum performance, security, and scalability before deploying it and also to make sure that the MongoDB database is configured and linked to the Node.js server

Step 11: Training and Support

To help users and administrators use the system effectively, provide training and support and provide contact details for assistance with technical issues.

Step 12: Continuous Improvement

Gather user and stakeholder feedback to make ongoing system improvements. To stay relevant and to meet evolving needs, the system should be updated as necessary and to maintain a sharp focus on security, privacy, and adherence to pertinent healthcare and data protection regulations throughout the development process. Work closely with healthcare professionals to make sure the system effectively satisfies the unique requirements of blood banks and hospitals.

IV Implementation And Results

The implementation can be understood using the flowchart as the process for managing blood donations and requests is depicted in the flowchart. The decision to donate blood is the first step. Donor registration and eligibility are checked to see if theyare eligible. The database is updated if they are qualified. Then, a patient can ask for blood. The recipient is informed if the blood is not available.

The information and the contact details from the database is retrieved and verified if the blood donor is available. If the data is accurate, it is shown to the recipient and the blood needer and they can contact them. In conclusion, the flowchart demonstrates the administration of blood donations and the satisfaction of recipient requests.

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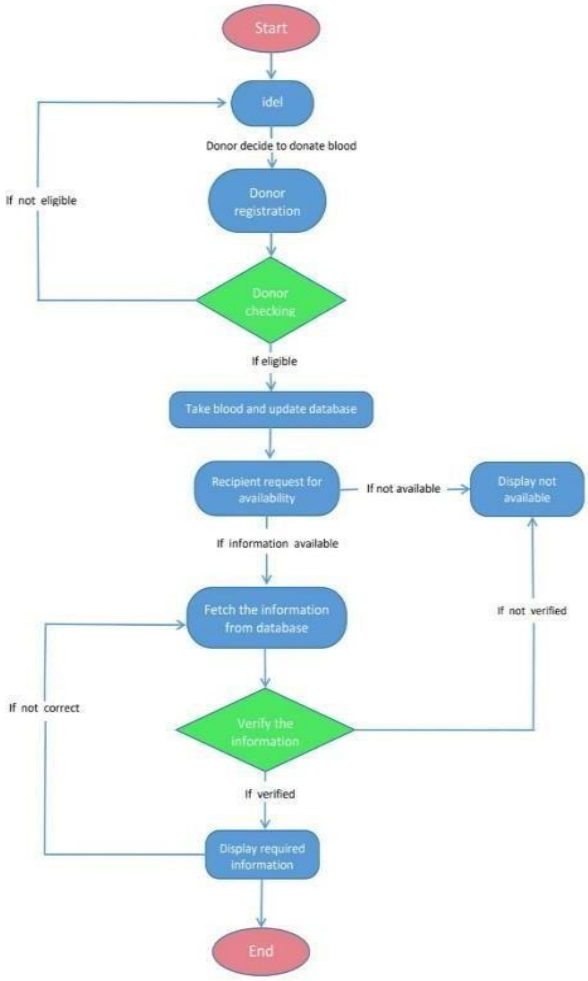

Figure 1 Flow Diagram

The outcomes can be categorized as shown in Figures 2 and 3. In which Figure 3 of the results displays the donor registrations with their name, gender, age, mobile no, blood group which is very useful to identify the details about the donor

Figure 2 Donar Registration

The project makes it easier for those in need to get the blood they need from different blood groups. Based on their location, The customer can learn more regarding nearby people who donate blood. The system makes all of the fundamental data that the donor needs available, making it easy for the user to use. The online blood supply administration system is a daily required and crucial system for a variety of people, including system workers, healthcare professionals, donors as well as people who received it, and other users.

During the construction of this project, we became aware of the manner in which life-saving blood donation is. This motivated and persuaded us to donate blood on a regular basis, as well as to motivate and persuade our fellow citizens to donate blood. A database has been established to store historical data related to blood donation and reception, important for most people to be aware of the advantages of such activities.

Figure 3 User Profile updation

The Figure 3 denotes that if user decides to update his profile(password) then we can select the user name and type the new password that we want and then select the type of user whether it is donor or blood needer or receptionist.

V CONCLUSION

Nearly every aspect of life, including business, healthcare, industry, and research settings, is gradually embracing computers. In these application areas, processes that save lives and provide healthcare are best handled by online systems. These facts can be manipulated by the suggested strategy. The study's primary goals were to use technology to create an online system for managing blood donations that would assist in keeping track of donors and locating a needed blood donor in an emergency. These analyses and reports show that it'sas well as data from camps, in order to make future decisions based on concrete analytical results.

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